

Multimodal imaging for Search and Rescue through UAVs

1. INTRODUCTION

The objective of this project is to build a deep learning model able to identify people in danger in a wild environment and recognize their position. The model should be multimodal and accept RGB or Thermal data.

Search-and-Rescue (SAR) operations aim to locate and rescue people in imminent danger, often in hostile environments like post-disaster areas, low light conditions, or remote locations. Speed is critical to prevent injuries or loss of life. Unmanned aerial vehicles (UAVs), or drones, are increasingly used to support these missions due to their cost-effectiveness and versatility [1–3]. UAVs face a unique set of algorithmic challenges in wilderness settings. To detect humans in such environments, the UAV must be capable of handling unstructured and rugged terrain, reasoning about occlusion caused by foliage and shadows, understanding the relative size of human objects from a high-altitude viewpoint, and maintaining robustness across diverse environments and weather conditions, such as snow, fog, forests, mountains, dawn, and dusk [4].

Under specific conditions, such as nighttime, thermal cameras can enhance detection performance. In contrast, RGB cameras are more effective when additional heat sources are present. Therefore, a versatile model capable of utilizing both modalities is essential.

2. METHOD

To accomplish this we will use the latest object detection deep learning model: YOLOv10 [5]. Datasets involved will be two: SARD [6] and WiSARD [4]. SARD (Search and Rescue Dataset) is an RGB object detection dataset of aerial scenes with people in danger in different poses. The poses which are the classes of the dataset are the following:

- Running
- Walking
- Laying down
- Not defined
- Seated
- Standing

This poses emulate people in danger in different situations. The dataset comprises 1,981 manually labeled RGB images from video sequences. WiSARD (Wilderness Search and Rescue Dataset) is a multimodal object detection dataset of aerial scenes with people in danger. This dataset provides a set of RGB sequences, a set of Thermal sequences and a set of synced RGB-Thermal sequences. Key features of the dataset include:

- Approximately 55,942 labeled images, including visual-only, thermal-only, and synchronized visual-thermal image pairs.
- Images captured in diverse wilderness environments like forests, fields, rocky areas, coastal regions, and snowy terrain.
- Data collected across different seasons, times of day (day, night, dawn, dusk), and weather conditions.
- Images taken from various altitudes (20-120m) and camera angles.
- Human subjects in different positions (sitting, standing, lying down) and actions (walking, running, waving).



Fig. S1. Tiling process to build the classification dataset. For each annotation, we extract a 224×224 patch and assign the corresponding pose label.

- Metadata tags for terrain type, time of day, human location, season, and camera view.
- Bounding box annotations for human subjects in the images.
- Designed to address challenges specific to wilderness search and rescue, such as occlusions, poor lighting, and thermal crossover effects.

The dataset aims to provide a benchmark for developing and testing computer vision algorithms, particularly object detection and multi-modal sensor fusion, in challenging wilderness search and rescue scenarios. The dataset is divided into three parts: visual-only, thermal-only, and synchronized visual-thermal image pairs.

Since the WiSARD dataset lacks pose labels, we need to transfer these labels using the SARD dataset. This involves first training a classifier on the SARD dataset to recognize and classify human poses in images. Once trained, the classifier is applied to the WiSARD dataset, enabling the identification of poses in its images. Thus, with the information both datasets we can build a multimodal model able to recognize the location and pose of people in danger.

A. Step 1: Train the pose classification model (ViT-B)

The first step is extracting the pose information from the SARD dataset to transfer it to WiSARD. We will use a state-of-the-art vision model: ViT-B (Vision Transformer Base) as an image classifier with 6 classes, each one corresponding to one pose. ViT-B is a transformer-based neural network architecture designed for image classification tasks, which divides images into fixed-size patches, linearly embeds them, and processes them with transformer encoder layers. This model has shown excellent performance on various computer vision tasks. Using the ground truth, we will build a classification dataset from each annotation, as is illustrated in S1.

B. Step 2: Labelling WiSARD (RGB / RGB - Thermal)

Once we obtained our trained ViT-B model, we can use it to label the RGB part of the WiSARD dataset. The WiSARD (Wilderness Search and Rescue Dataset) is a large-scale dataset of labeled visual and thermal images collected from UAVs for wilderness search and rescue applications.

The process in S1 is repeated on WiSARD and the trained model is applied to this dataset. Note that labelling all the RGB images, we will also automatically label also the Thermal ones coupled with a RGB image.

C. Step 3: Training YOLOv10

Using the labelled RGB / RGB-Thermal dataset, we can now train the multimodal YOLOv10 model. This model is a slightly modified version; to accommodate the possibility of the two optional inputs we designed a new module: the Optional Convolution. When the input is supplied it acts as a traditional convolution, when it is not supplied, it outputs an embedding that informs the subsequent layer the absence of the image:

$$x_{out} = \begin{cases} \text{Conv}(x_{RGB}) + \text{Conv}(x_{IR}), & \text{if } x \text{ is supplied,} \\ \text{Conv}(x_{RGB}) + E_{IR}, & \text{if } x_{IR} \text{ is not supplied,} \\ \text{Conv}(x_{IR}) + E_{RGB}, & \text{if } x_{RGB} \text{ is not supplied,} \end{cases}$$

where E_{RGB} and E_{IR} are the learnable embeddings which handle the missing image. Conv is a standard convolutional layer. The architecture is visualized S2.

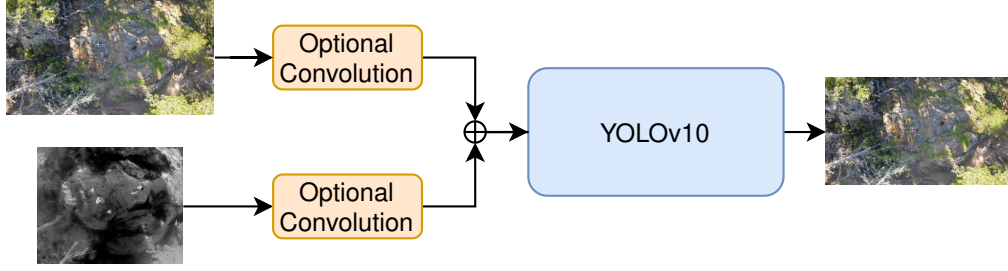


Fig. S2. Fusion YOLOv10, based on YOLOv10. Using OptionalConvolutions the model can achieve flexibility in relation to the given data.

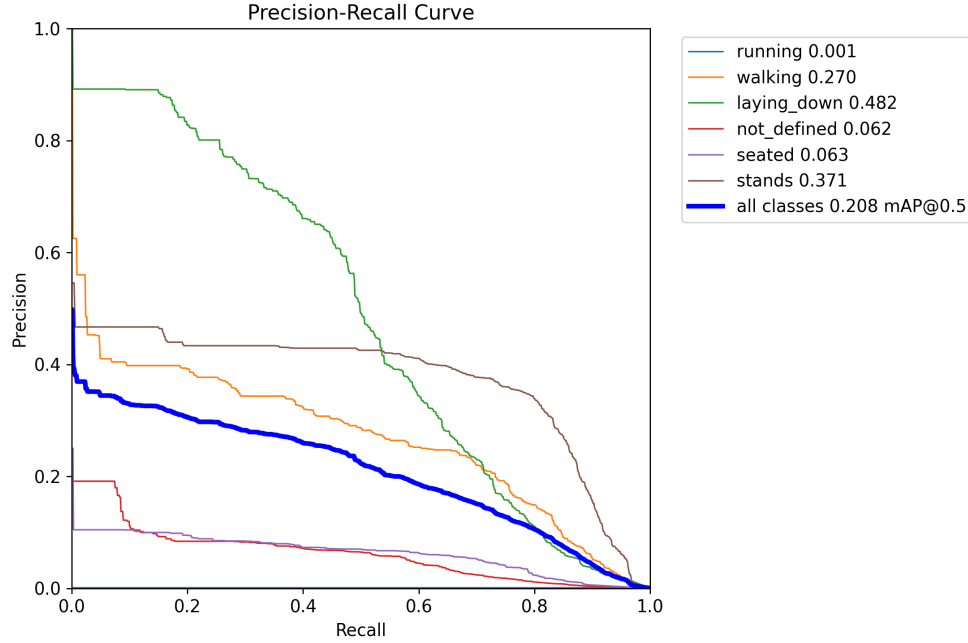


Fig. S3. Precision-Recall curve of YOLOv10-Small on WiSARD RGB part

As a first experiment, we trained the standard YOLOv10 on the RGB part of the WiSARD dataset. The version used was the YOLOv10-Small. The model was trained for 50 epochs with a batch size of 16 and a learning rate of $1.0e-5$, given that we are fine-tuning the model. The precision-recall curve is shown in S3.

Given the unsatisfactory results, we tried to reduce the number of classes by merging the classes that are similar. The new classes are the following:

- Stands (stands, walking, running);
- Rests (seated, laying down);
- Not defined.

The training and test was repeated with the new classes. The precision-recall curve is shown in S4. The mAP@0.5 increased from 0.208 to 0.371, which is a natural improvement due to the reduction of classes. However, the model still struggles to detect people in the images. We tried to further reduce the number of classes to one, but the results were not satisfactory, as shown in S5. Therefore, we decided to keep only one class and don't consider the pose anymore. The error from the model trained on the SARD dataset propagates to the WiSARD dataset making the model unable to detect the people in the images.

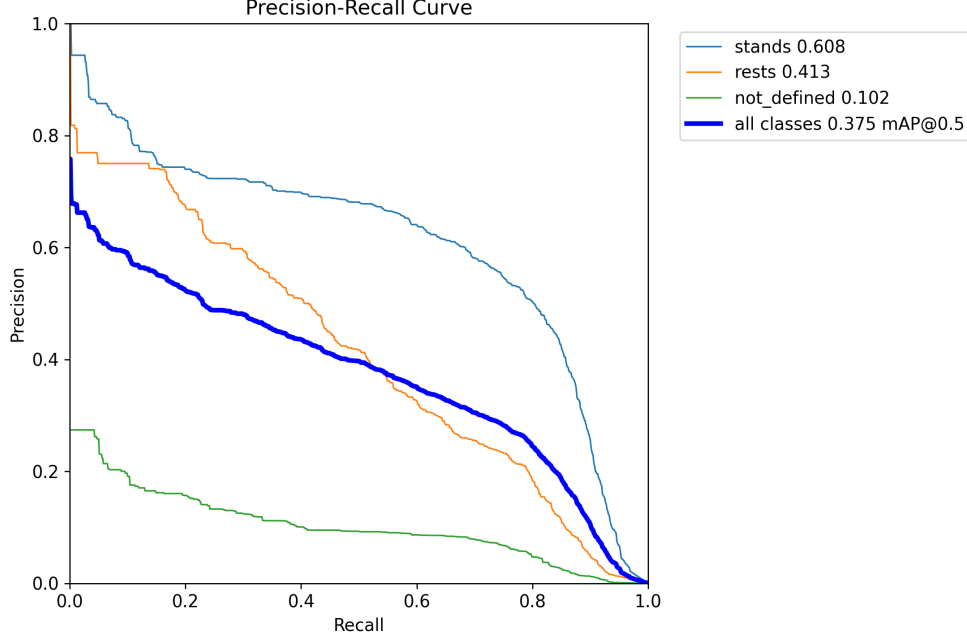


Fig. S4. Precision-Recall curve of YOLOv10-Small on WiSARD RGB part with reduced classes.

WiSARD dataset contains very small objects, compared to the usual detection datasets, indeed, authors of the dataset develop trained YOLOv5 on 512×512 tiles of the dataset, instead of the whole image, in order to avoid this problem.

The WiSARD dataset presents a unique challenge in object detection due to the extremely small size of the human subjects in the images, especially when compared to conventional detection datasets. This characteristic stems from the aerial nature of the data collection, where UAVs capture images from significant altitudes.

3. FUSION DETR

The WiSARD dataset presents a significant challenge in object detection due to misaligned RGB-IR (Red-Green-Blue and Thermal) images, which renders traditional CNN-based models like YOLO ineffective. We propose adapting DETR (DEtection TRansformer), a transformer-based object detection model, to address this challenge.

DETR uses a CNN to extract a feature map $F \in \mathbb{R}^{H \times W \times C}$ from an image $I \in \mathbb{R}^{H \times W \times 3}$, where H and W are the height and width of the image, respectively, and C is the number of channels. Each pixel in the feature map is a vector of length C and will be used as a query in the transformer. The transformer decoder will use cross-attention between a learnable set of tokens and the feature map to predict the bounding boxes and class labels for each object in the image.

The main idea is to add a new set of tokens for the IR images extracted from another CNN, in this way even if the images are not aligned, the DETR model will be able to detect and localize the people in the images thanks to the attention mechanism that will attend all the tokens of the IR images and the feature map of the RGB images. The model is visualized in S6. The Cross-modal fusion is achieved by concatenating RGB and IR tokens:

$$T_{\text{combined}} = \text{Concat}(T_{\text{RGB}}, T_{\text{IR}}) \quad (\text{S1})$$

The proposed approach transforms the challenge of image misalignment into an opportunity for adaptive, context-aware feature integration. By extracting tokens from both RGB and IR images and applying the transformer’s cross-attention mechanism, we enable robust object detection that can dynamically correlate features across modalities, effectively compensating for geometric misalignments. The model’s architecture provides flexibility for RGB-only, IR-only, or multimodal RGB-IR detection, addressing the complex detection requirements of the WiSARD dataset.

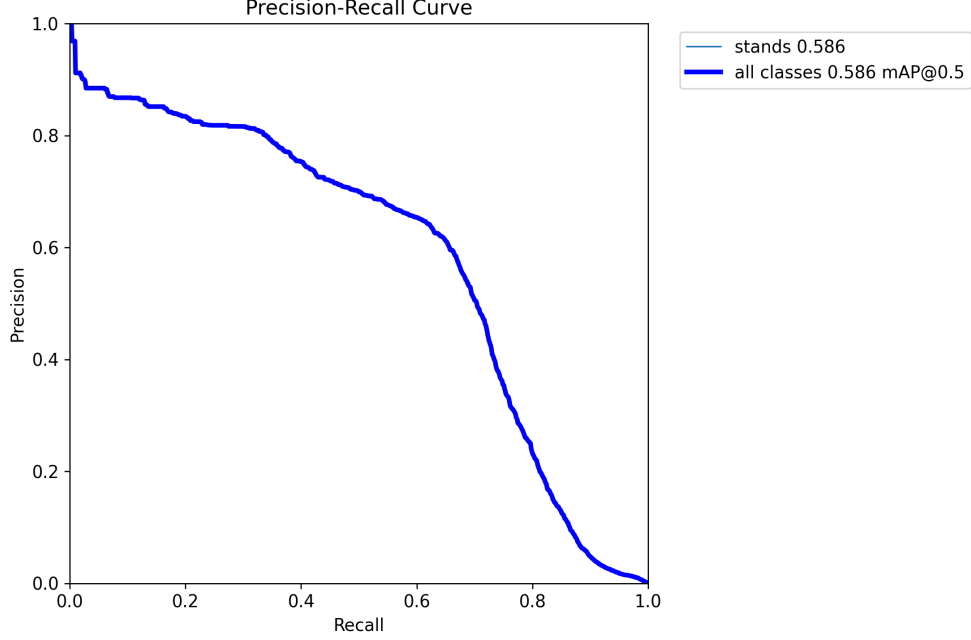


Fig. S5. Precision-Recall curve of YOLOv10-Small on WiSARD RGB part with one class.

4. EXPERIMENT

The experimental setting consist in the separation of the RGB and IR parts of the WiSARD dataset, to test the effectiveness of the fusion model. The whole can be described as $D = D_{RGB} \cup D_{IR} \cup D_{RGB-IR}$ where D_{RGB} , D_{IR} and D_{RGB-IR} are the RGB, IR and RGB-IR parts of the dataset, respectively. For our experiments we rearranged the dataset in the following way:

1. **Full RGB only part** (D_{RGB_o}):

$$D_{RGB_o} = D_{RGB} \cup \{x_{RGB} \mid \forall (x_{RGB}, x_{IR}) \in D_{RGB-IR}\}$$

2. **Full IR only Part** (D_{IR_o}):

$$D_{IR_o} = D_{IR} \cup \{x_{IR} \mid \forall (x_{RGB}, x_{IR}) \in D_{RGB-IR}\}$$

3. **Full RGB only Part + IR Synced Part** ($D_{RGB_o-IR_s}$):

$$D_{RGB_o-IR_s} = D_{RGB-IR} \cup D_{RGB}$$

4. **Full IR only Part + Synced RGB Part** ($D_{RGB_s-IR_o}$):

$$D_{RGB_s-IR_o} = D_{RGB-IR} \cup D_{IR}$$

5. **RGB-IR Synced Part:**

$$D_{RGB-IR}$$

6. **Whole Dataset:**

$$D$$

The first two subsets contain images in only one modality: training the model on them will result in RGB-only or IR-only detection, thus it is equivalent to train the standard DETR model. The third and four subsets, have respectively the same number of elements of the first two subsets, allowing for comparison of the fusion model with the standard DETR model. The fifth subset contains only the RGB-IR synced part of the dataset while the last one is the whole dataset. Each of these subsets are divided into a training set (10 folders - 21029 samples), a validation set (6

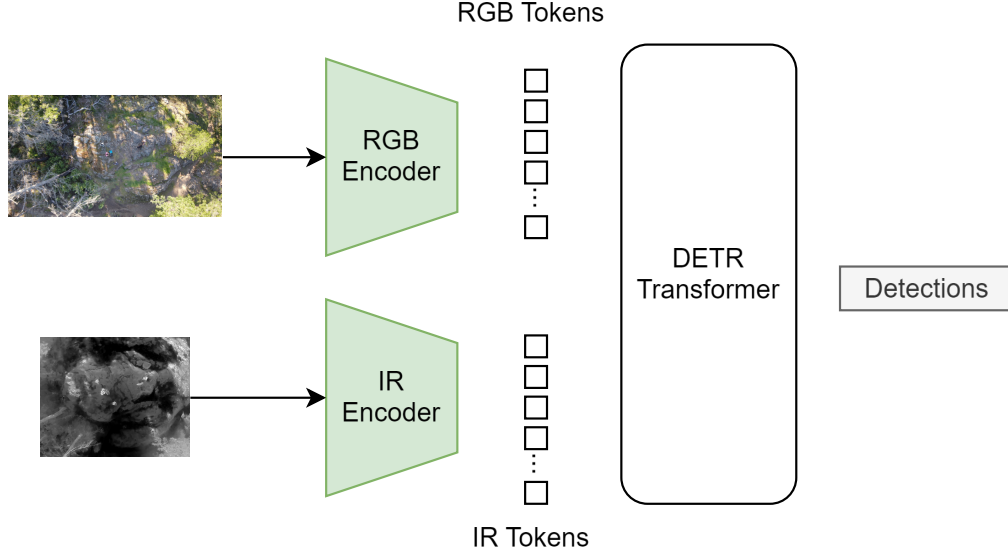


Fig. S6. Fusion DETR. The model uses two encoders that extract features from the RGB and IR images. The model then uses a transformer decoder to predict the bounding boxes and class labels for each object in the image.

folders - 4824 samples) and a test set (7 folders - 5591 samples). The training set is used to train the model, the validation set is used for early stopping and the test set is used for evaluation. Unlike the experiments from the dataset authors, the splits ensure that all the frames of the same video are in the same set, so that the same video is not used for training and testing.

We used as evaluation metric the $mAP@50$ IoU metric, which is the average precision at 50% IoU threshold. We avoid using the standard $mAP : 0.5 : 0.95$ metric because in this context the IoU threshold is not relevant as we are interested in the detection of people and not in an accurate localization of them. Also given that the targets are small objects, the IoU threshold of 50% is a suitable choice.

The results are presented in Table S1. When tested on the entire dataset, as expected, the best performance is achieved by training on the entire dataset. For the IR portion, training exclusively on the IR subset yields the best results, indicating that a specialized model is more effective for IR-only images. Conversely, for the RGB portion, the highest performance is obtained by training on the entire dataset. This suggests that incorporating IR images can enhance the performance for RGB data. Additionally, the model trained on the combined RGB + IR Synced subset outperforms the model trained solely on the RGB subset. This finding highlights that even with the same number of samples, including an IR-synced component leads to improved results.

5. CONCLUSIONS

This study focused on developing a deep learning framework to identify individuals in danger in wilderness environments using a multimodal approach. Key findings include:

- **Pose Classification and Label Transfer:** The transfer of pose labels from the SARD dataset to the WiSARD dataset using the Vision Transformer Base (ViT-B) was unsuccessful. Errors in label transfer significantly impacted detection performance, leading to the decision to simplify to a single detection class.
- **Challenges with YOLOv10:** YOLOv10 struggled due to misalignments in RGB-IR data and the inherently small object sizes in the dataset. These issues, coupled with the limitations of training on smaller tiles, resulted in suboptimal performance.
- **Fusion DETR Effectiveness:** The proposed Fusion DETR model successfully integrated RGB and thermal data using a transformer-based architecture. Its cross-attention mechanisms allowed robust detection despite geometric misalignments.

Trained on	Tested on	mAP@50
$D_{RGB_a-IR_s}$	D	0.263
D	D	0.372
D_{RGB-IR}	D	0.151
$D_{RGB_s-IR_a}$	D	0.249
$D_{RGB_a-IR_s}$	D_{IR_o}	0.0
D	D_{IR_o}	0.233
D_{IR_o}	D_{IR_o}	0.427
D_{RGB-IR}	D_{IR_o}	0.0
$D_{RGB_s-IR_a}$	D_{IR_o}	0.292
$D_{RGB_a-IR_s}$	D_{RGB_o}	0.442
D	D_{RGB_o}	0.465
D_{RGB-IR}	D_{RGB_o}	0.281
D_{RGB_o}	D_{RGB_o}	0.432
$D_{RGB_s-IR_a}$	D_{RGB_o}	0.188

Table S1. Results on the various subsets by varying the training set

- **Experimental Insights:** Training on the entire dataset provided the best overall performance. Specialized training improved results for single modalities, and synchronized RGB-IR data enhanced RGB-only detection.

Future work should prioritize improving RGB-IR alignment, refining label transfer methods, and exploring advanced transformer architectures for robust and efficient multimodal detection in UAV applications.

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